TASK SPECIFICATION: P6-06 PHASE P6 — ACADEMIC CAMPUS SERVICES & RESOURCE PODS

Study Rooms & Computer Labs Reservation API

Git Branch: `feature/P6-06-rooms-api` · Target Pull Request: feature/P6-06-rooms-api

Core Objective & Technical Scope:
Collaborative study pods and laboratory booking engine with active session extension, release, and GiST overlap prevention.

Priority Level	High	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P6-06-rooms-api`
Upstream Dependencies	P2-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V10__rooms.sql migration with btree_gist exclusion on room booking intervals.
2. Implement Room and RoomBooking domain entities.
3. Implement RoomService supporting create, extend, and release operations.
4. Enforce maximum consecutive booking duration limits.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V10__rooms.sql`
- `rooms/**`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Collision-free scheduling guaranteed across all study pods and labs.
- Early release immediately frees slot for other waiting workgroups.

VERIFICATION & TEST COVERAGE STRATEGY

- RoomServiceTest
- RoomBookingExclusionTest
- RoomControllerTest